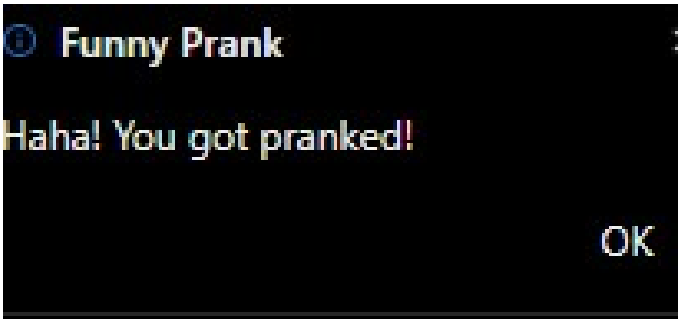
**1.Create a simple Python script named prank_message.py that, when run, repeatedly opens a pop-up message box with a fun message until the user closes the program.

Hint: Use the tkinter library's messagebox in a loop.**

```
import tkinter as tk
from tkinter import messagebox

root = tk.Tk()
root.withdraw()

while True:
    messagebox.showinfo(
        "Funny Prank",
        "Haha! You got pranked!"
    )
```



**2.Write a JavaScript snippet that, when pasted into the browser console, automatically opens new tabs with a chosen URL (such as your favorite meme page) every 2 seconds until stopped.

Constraint: Limit the total number of tabs opened to 5 to avoid browser overload.**

```
let count = 0;
```

```
let timer = setInterval(() => {
   if (count >= 5) {
      clearInterval(timer);
      console.log("Stopped opening tabs");
      return;
   }

   window.open("https://example.com", "_blank");
   count++;

   console.log("Tab opened:", count);
}, 2000);
```

```
Tab opened: 1
Tab opened: 2
Tab opened: 3
Tab opened: 4
Tab opened: 5
Stopped opening tabs
```

**3.Simulate a worm-like behavior by creating a Python script called file_spreader.py that copies itself into every folder inside a given directory path.

Hint: Use os.listdir() and shutil.copy(). Do not actually spread harmful files—use a harmless text file or your script as the payload.**

```
import os
import shutil
```

```
folder = "test_folder"

os.makedirs(folder, exist_ok=True)

# Create three test folders
for i in range(1, 4):
    path = os.path.join(folder, f"folder{i}")
    os.makedirs(path, exist_ok=True)

# Create a harmless text file
with open("demo.txt", "w") as file:
    file.write("This is a harmless worm demo.")

# Copy the file into test folders
for name in os.listdir(folder):
    path = os.path.join(folder, name)

    if os.path.isdir(path):
        shutil.copy("demo.txt", path)
        print("Copied into:", name)

print("Simulation completed!")
```

```
Copied into: folder1
Copied into: folder2
Copied into: folder3
Simulation completed!
```

4.Modify the prank_message.py script so that it creates a log file named worm_log.txt, recording the date and time each time the pop-up appears.

```
import tkinter as tk
from tkinter import messagebox
from datetime import datetime

root = tk.Tk()
root.withdraw()

while True:
    now = datetime.now()

    with open("worm_log.txt", "a") as file:
        file.write(
            f"Popup shown at: {now}\n"
        )

    messagebox.showinfo(
        "Funny Prank",
        "Haha! You got pranked!"
    )
```



5.Research using ChatGPT: Ask ChatGPT to suggest 3 ways a basic computer worm could be detected by antivirus software, and write a short explanation of each method in your own words.

Detection Method	Easy Explanation
1. Signature-Based Detection	Antivirus compares files with known malware patterns.
2. Behavior-Based Detection	Antivirus checks unusual activities, such as files copying themselves repeatedly.
3. Network-Based Detection	Antivirus or network security tools detect unusual connections and rapid spreading between computers.

Explanation

1. Signature-Based Detection

Antivirus software checks files for known virus or worm signatures. If a matching signature is found, it can identify the malware.

2. Behavior-Based Detection

Antivirus monitors suspicious activities, such as a program copying itself into many folders or running without permission.

3. Network-Based Detection

Security software monitors network traffic. It can detect worms trying to spread to other computers.

Result: We learned three methods used to detect computer worms.